Heat Diffusion Equation

Class Assignment

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1 Question

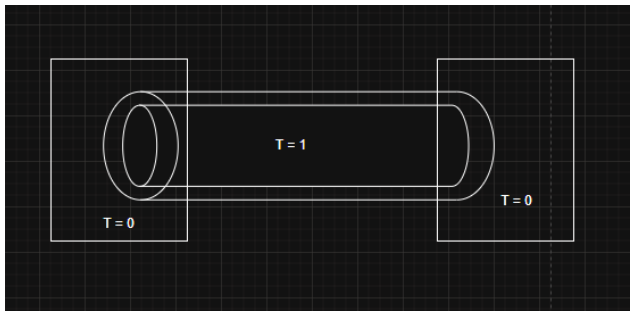


Figure 1: Quesiton fig

2 Results

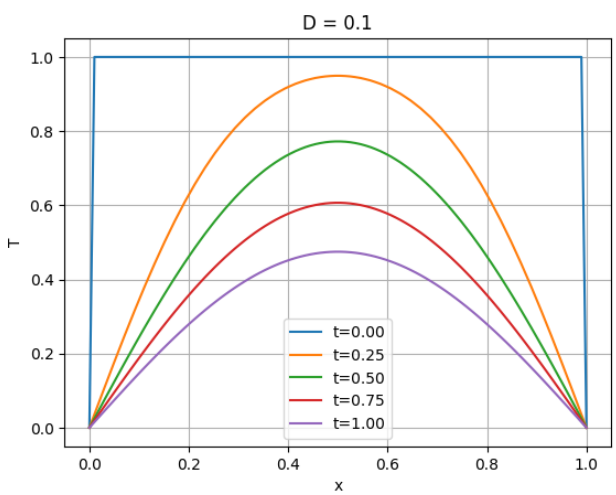
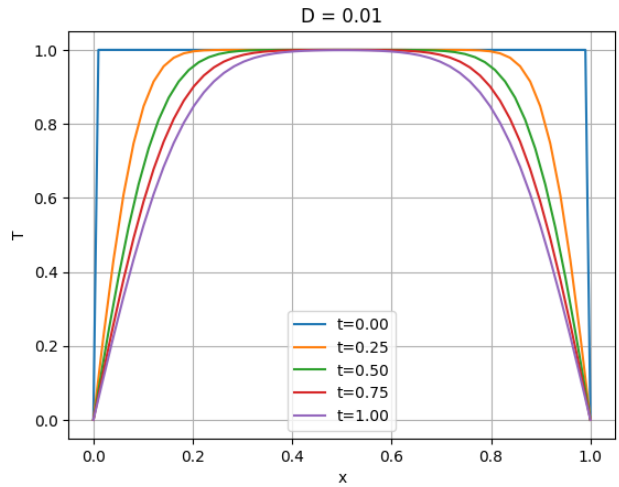
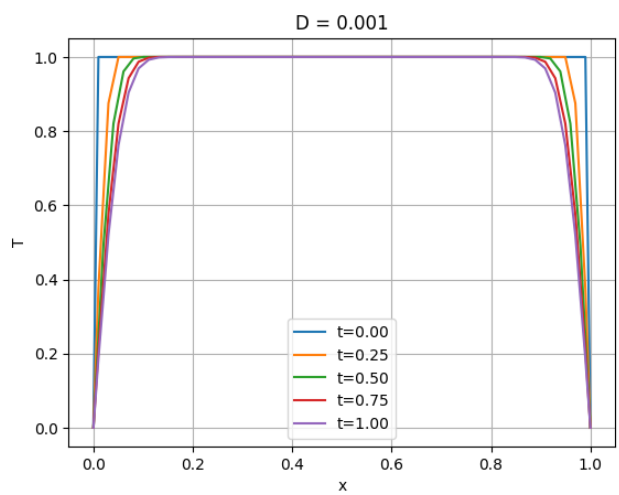


Figure 2





The heat equation describes how temperature changes with time inside a material. It states that heat naturally flows from regions of higher temperature to lower temperature, and the rate of change depends on how curved the temperature distribution is. The constant D represents how quickly heat diffuses through the material. To solve the equation, we need initial conditions (temperature at the start) and boundary conditions (temperature or heat flow at the ends). As time progresses, the temperature distribution becomes smoother and eventually reaches a steady state where no further changes occur.

A Python Code

```
import numpy as np
import matplotlib.pyplot as plt

#parameters
xi, xf = 0, 1
```

```
D = 0.1

#grid
x = np.linspace(xi, xf, 100)
dx = x[1] - x[0]
dt = 0.5 * dx**2 / D
nt = int(1 / dt)

#temperature grid
T = np.zeros((len(x), nt))

#initial condition
T[1:-1,0] = 1


#time evolution
for j in range(nt - 1):
    for i in range(1, len(x) - 1):
        T[i, j+1] = T[i, j] + (D * dt / dx**2) * (
            T[i+1, j] - 2*T[i, j] + T[i-1, j]
        )

# Plot
plt.figure()
plt.plot(x, T[:, 0], label='t=0.00')
plt.plot(x, T[:, int(0.25*nt)], label='t=0.25')
plt.plot(x, T[:, int(0.5*nt)], label='t=0.50')
plt.plot(x, T[:, int(0.75*nt)], label='t=0.75')
plt.plot(x, T[:, -1], label='t=1.00')
plt.title("D = 0.1")
plt.xlabel("x")
plt.ylabel("T")
plt.legend()
plt.grid()
plt.show()
```